

TIC 467962259

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 4.061582 d
Epoch (t0) 1657.6207 BTJD
Duration 0.1200 d
Depth 3.893 % (38,933 ppm)
SNR 48.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

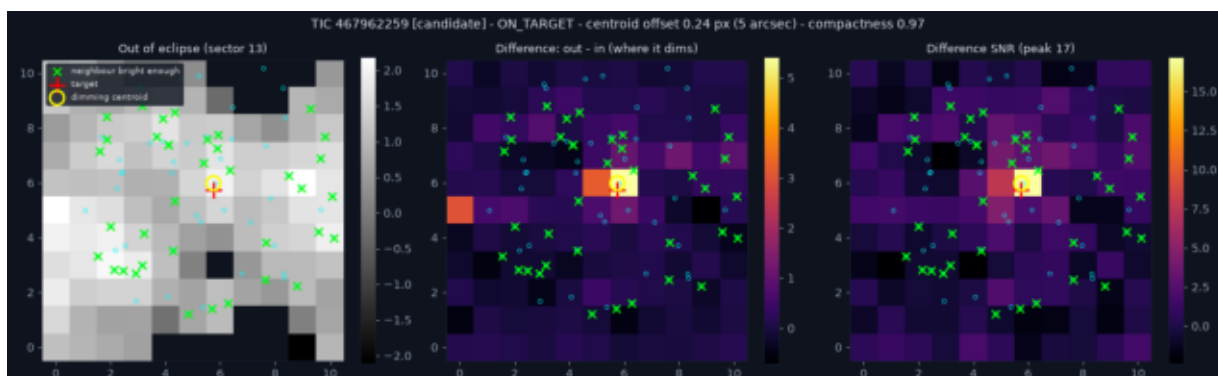
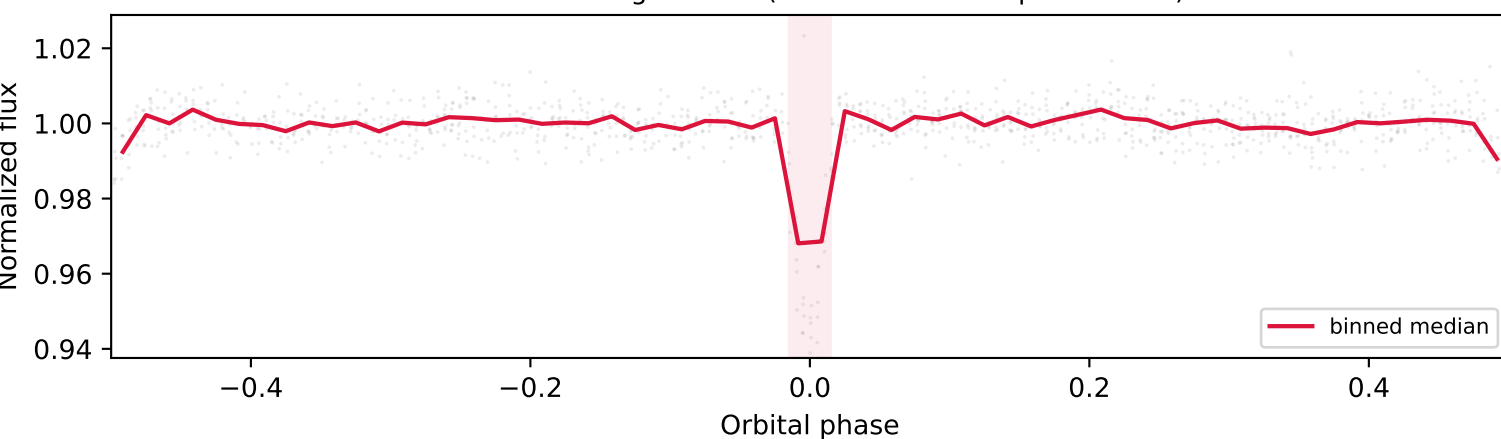
Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.00 (gate ≥ 0.50)
Robust depth sigma 37.9 (gate ≥ 7.0)
Odd/even z-score 0.40 (consistent)
Secondary eclipse detected, phase 0.49, 5.0 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 40.6 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 17.22 (gate ≥ 5.0) Compactness 0.97 (gate ≥ 0.50)
Dimming centroid offset 0.24 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 467962264
0.76 px away, dTmag +0.64 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.